

# Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

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*Why Software Tests Fail and Hardware Attestation Wins*

[arxiv.org/abs/2504.04715](https://arxiv.org/abs/2504.04715)



# The Model Substitution Problem

*Economic Incentives Drive Covert Swaps*

Commercial LLM API providers charge for specific models but the black-box nature of APIs means users have no guarantee the advertised model is actually being served.

1

## Smaller Model

8B instead of 70B

2

## Quantized Variant

FP8/INT8 instead of FP16

3

## Fine-tuned Version

Updated or retrained model

4

## Different Family

Entirely different architecture

**Formal Framework:**  $H_0: P(y|x) = P_{\text{spec}}(y|x)$  vs  $H_1: P(y|x) = P_{\text{actual}}(y|x)$

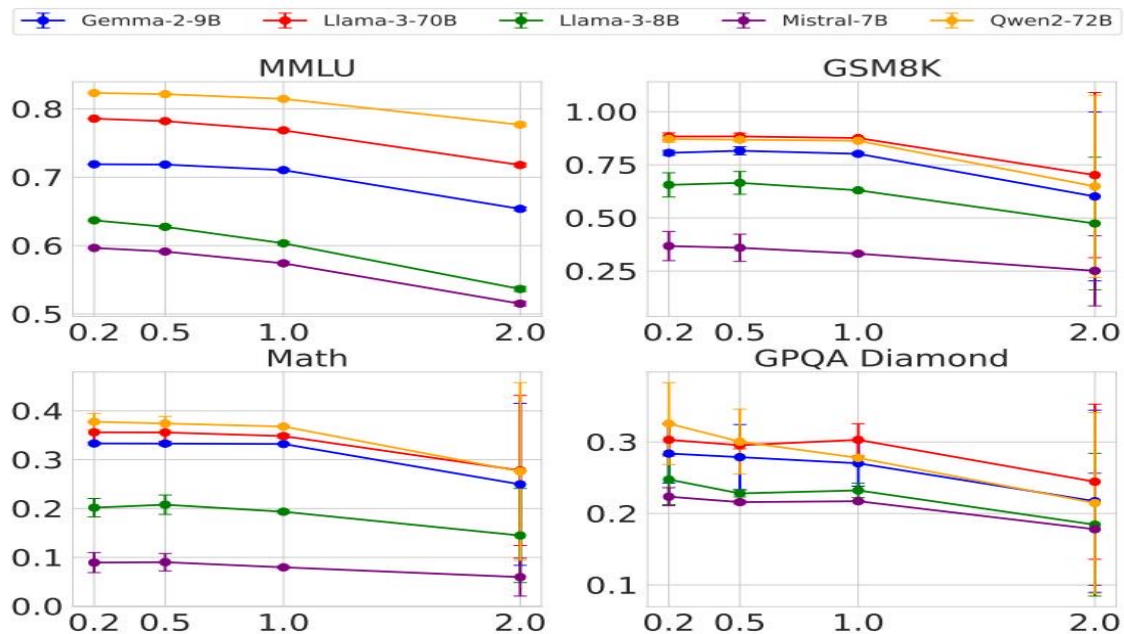
# FP8 Quantization Preserves Most Accuracy

*Making Substitution Hard to Detect*

| Model                           | MMLU         | GSM8K        | MATH         | GPQA Diamond |
|---------------------------------|--------------|--------------|--------------|--------------|
| <b>Llama-3-8B-Instruct</b>      | 62.69 ± 0.18 | 61.14 ± 3.47 | 20.65 ± 3.43 | 22.62 ± 0.22 |
| → FP8                           | 62.43 ± 0.26 | 60.90 ± 4.10 | 14.91 ± 2.49 | 20.14 ± 0.24 |
| <b>Llama-3-70B-Instruct</b>     | 78.05 ± 0.08 | 88.06 ± 1.44 | 35.69 ± 1.33 | 29.60 ± 0.51 |
| → FP8                           | 77.88 ± 0.13 | 87.35 ± 1.24 | 35.75 ± 1.16 | 33.12 ± 0.30 |
| <b>Gemma-2-9b-it</b>            | 71.86 ± 0.08 | 81.80 ± 1.35 | 33.41 ± 0.28 | 28.64 ± 2.97 |
| → FP8                           | 71.92 ± 0.11 | 79.41 ± 1.14 | 32.53 ± 0.34 | 27.81 ± 3.22 |
| <b>Qwen2-72B-Instruct</b>       | 82.18 ± 0.08 | 86.72 ± 1.00 | 37.39 ± 1.41 | 29.93 ± 2.71 |
| → FP8                           | 81.98 ± 0.08 | 86.82 ± 0.97 | 37.67 ± 1.39 | 31.08 ± 1.96 |
| <b>Mistral-7B-Instruct-v0.3</b> | 59.15 ± 0.10 | 35.90 ± 4.54 | 8.94 ± 1.24  | 21.60 ± 0.17 |
| → FP8                           | 58.77 ± 0.13 | 32.20 ± 4.02 | 7.68 ± 1.23  | 22.72 ± 0.19 |

**Key Insight:** For most models, FP8 scores fall within the error bars of the original, especially on MMLU and GSM8K. Quantization saves significant GPU costs while being nearly undetectable via benchmark comparison.

# Higher Temperature Amplifies Gaps but Increases Noise



## Key Observations

**Temperature 0.2-0.5:** Performance gaps between original and FP8 models are small

**Temperature 1.0:** All models degrade; substitution signal weakens

**Temperature 2.0:** Severe degradation across all models; gap narrows or becomes noisy, complicating statistical detection

**Implication:** Higher temperatures do not reliably amplify the substitution signal -- they increase variance, making statistical detection even harder.

# Log Probability Fingerprinting Fails

*Due to Inference Nondeterminism*

| Model                       | BERT Acc | T5 Acc | GPT2 Acc | LLM2Vec Acc |
|-----------------------------|----------|--------|----------|-------------|
| Llama3-70B-Instruct-FP8     | 50.55    | 50.10  | 50.45    | 49.90       |
| Llama3-70B-Instruct-INT8    | 51.60    | 49.90  | 51.30    | 50.25       |
| Gemma2-9b-it-FP8            | 49.95    | 50.30  | 51.20    | 49.80       |
| Gemma2-9b-it-INT8           | 49.00    | 49.70  | 51.65    | 49.55       |
| Mistral-7b-v3-Instruct-FP8  | 50.55    | 49.75  | 48.70    | 48.75       |
| Mistral-7b-v3-Instruct-INT8 | 49.70    | 50.75  | 50.50    | 51.15       |
| Qwen2-72B-Instruct-FP8      | 50.05    | 50.25  | 48.75    | 49.80       |
| Qwen2-72B-Instruct-INT8     | 50.75    | 50.55  | 49.45    | 50.20       |

## Sources of Variation

### Software Frameworks

transformers 4.40 vs 4.50vLLM 0.6.2 vs 0.8.2

### GPU Hardware

NVIDIA A100 vs H100

**Result:** Cannot distinguish legitimate infrastructure differences from actual substitution

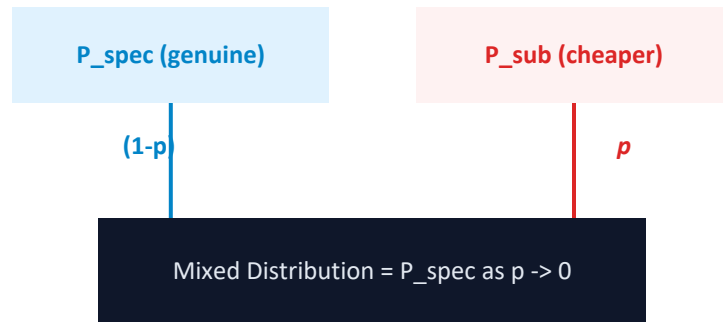
**Core Problem:** Production-level nondeterminism defeats metadata-based detection. An auditor cannot tell if log-prob differences come from a substituted model or simply a different inference stack.

# Randomized Routing Makes Detection Query-Prohibitive

## The Attack

Provider routes only a fraction  $p$  of queries to the cheaper substitute model. The remaining  $(1-p)$  queries go to the genuine model.

As  $p \rightarrow 0$ , the mixed distribution **converges to  $P_{\text{spec}}$** , requiring exponentially more queries to detect the substitution.



## Adversarial Strategy: Benchmark Evasion

Provider detects known benchmark prompts and routes them to the genuine model while serving substitutes for regular traffic. Audits using standard benchmarks are trivially evaded.

**Combined Effect:** Randomized substitution + benchmark evasion make software-only detection fundamentally query-prohibitive, even with infinite query budgets.

# Software-Only Auditing Methods Are Fundamentally Unreliable

| Method Type                                                                      | Core Weakness                                                                                   | Practical Limitation                                                                                                       |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Text-Based Statistical Tests</b><br>(e.g., MMD)                               | Requires impractically large query budgets; fails against subtle substitutions                  | Even with millions of queries, cannot reliably distinguish FP8 from FP16 when differences fall within error bars           |
| <b>Log-Probability Fingerprinting</b>                                            | Defeated by production-level inference nondeterminism across software versions and GPU hardware | Cannot distinguish legitimate infrastructure differences from actual model substitution                                    |
| <b>Adversarial Countermeasures</b><br>(Randomized routing,<br>Benchmark evasion) | Providers actively evade audits by mixing substitute responses or detecting benchmark prompts   | As $p \rightarrow 0$ , mixed distribution converges to $P_{\text{spec}}$ ; statistical detection becomes query-prohibitive |

**Verdict:** No software-only method can provide reliable model integrity guarantees. Each approach has fundamental blind spots that adversarial providers can exploit.

# Trusted Execution Environments

## *Provable Cryptographic Model Integrity*

### How TEEs Work

Hardware-secured enclaves provide cryptographic attestation of the code and data being executed.

- Prove specific model weights are loaded
- Verify correct inference code is running
- Only modest performance overhead

### TEE vs. Software Methods

Provable

Cryptographic proof

Statistical estimates

Efficient

Modest overhead

Query-prohibitive

Robust

Adversary-proof

Easily evaded

**Practical Deployment:** NVIDIA Confidential Computing on H100 GPUs enables TEE-based attestation today. Unlike Zero-Knowledge Proofs (ZKPs), which are computationally prohibitive for large models, TEEs are practical and deployable now.



# Key Takeaways

## *Close the Verification Gap with Hardware Attestation*

- 01 | Model substitution is economically motivated and practically undetectable via software methods alone.
- 02 | FP8 quantization barely affects benchmark scores — performance drops often fall within error bars.
- 03 | Inference nondeterminism across frameworks and hardware kills log-probability fingerprinting.
- 04 | Randomized substitution and benchmark evasion make statistical detection query-prohibitive.
- 05 | Trusted Execution Environments are the only currently viable path to verifiable LLM APIs.

**Call to Action:** The community must adopt hardware-based attestation as the standard for LLM API integrity verification.